CLAUDE CERTIFIED ARCHITECT — PROFESSIONAL

Objectives Index

All 38 official task-area objectives for *Claude Certified Architect – Professional (CCAR-P)*, each with its own page of certification-style practice questions (15 base questions each, plus supplementary questions added after cross-referencing other independently published CCAR-P study materials to close topic gaps — see the main README).

#	Domain	Objective	Page
1	Solution Design & Architecture	Translate a stated business problem into a Claude-based solution by identifying the actual underlying problem, defining the users, inputs, outputs and expected outcomes, distinguishing which subtasks genuinely require an LLM versus deterministic software, establishing measurable success criteria and business KPIs, and assessing feasibility, risk and organizational constraints before committing to an architecture.	obj-01
2	Solution Design & Architecture	Design complete end-to-end architectures that trace the full lifecycle from input through processing to output and feedback, clearly identifying where Claude, backend services and human reviewers each participate, incorporating validation and error-handling paths at every stage, and closing the loop with feedback mechanisms that improve future system performance.	obj-02
3	Solution Design & Architecture	Select the most appropriate architectural pattern for a given problem among augmented LLM, prompt chaining, routing, parallelization, orchestrator-workers, evaluator-optimizer and autonomous agent architectures, and correctly judge when a fixed workflow is sufficient versus when an autonomous agent is warranted.	obj-03
4	Solution Design & Architecture	Design multi-agent systems using coordinator and subagent architectures, including task delegation and decomposition, choices between parallel and sequential execution, context passing between agents, clearly scoped agent responsibilities and tool access, management of token consumption, concurrency and execution limits, and handling of failed or incomplete subagent runs.	obj-04
5	Solution Design & Architecture	Apply task decomposition techniques by breaking complex tasks into independent subtasks, identifying dependencies between them, defining clear contracts between components, ensuring	obj-05

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		each step is independently testable, and deciding which subtasks should run in parallel versus sequentially.	
6	Solution Design & Architecture	Align Claude-based architecture decisions with measurable business value, including efficiency gains, productivity improvements, broader business transformation, cost reduction, performance against service-level agreements, and demonstrable return on investment tied to business impact.	obj-06
7	Claude Models, Prompting & Context Engineering	Select Claude models based on capability, latency, and cost trade-offs, including reasoning/effort settings, migration strategy, and model assignment across stages of an agentic workflow.	obj-07
8	Claude Models, Prompting & Context Engineering	Design system prompts, reusable templates, and guardrails, including role/instruction separation, variable handling, output specifications, prompt versioning, and separating trusted instructions from untrusted user content.	obj-08
9	Claude Models, Prompting & Context Engineering	Apply prompting techniques—zero-shot, few-shot, chain-of-thought/extended reasoning, example selection, explicit instructions, and structured output—choosing the right technique and knowing when examples outperform additional instructions.	obj-09
10	Claude Models, Prompting & Context Engineering	Optimize context-window usage and token budgets through prioritization, conversation-history management, summarization/compaction, retrieval-based access to documents, tool-result optimization, subagent context isolation, and long-running session management.	obj-10
11	Claude Models, Prompting & Context Engineering	Implement prompt reuse through prompt caching, modular prompt architecture, reusable components, version management, Claude Skills with progressive loading, and the separation of static versus dynamic context.	obj-11
12	Integration	Evaluate tool and agent configurations to identify and remediate capability bloat, including tool selection and distribution, tool definitions and descriptions, redundant tooling, tool grouping and naming, limiting agent capabilities, reducing unnecessary context consumption, and progressive tool discovery.	obj-12
13	Integration	Analyse authentication and authorisation requirements for Claude-based integrations, including the distinction between authentication and authorisation, OAuth 2.1 fundamentals, API keys and credentials, delegated user identity, token handling and audience validation, least-privilege permissions, role-based	obj-13

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		access control, tenant isolation, secure secrets management, and preventing privilege escalation.	
14	Integration	Evaluate trade-offs between accuracy and latency in Claude-based systems, including model selection, reasoning effort, retrieval depth, reranking, validation and retries, parallel tool execution, streaming, prompt caching, and performance benchmarking.	obj-14
15	Integration	Analyse observability challenges in Claude-based agentic systems, including distributed tracing, structured logging, metrics and dashboards, correlation IDs, tool execution traces, agent execution monitoring, token consumption tracking, latency percentiles, error rates and failure monitoring, and privacy-aware logging.	obj-15
16	Integration	Design retrieval-augmented generation (RAG) pipelines for Claude-based systems, including document ingestion, chunking strategies, chunk size and overlap, contextual retrieval, embeddings, vector indexing, metadata filtering, document permissions, retrieval quality, and reranking.	obj-16
17	Integration	Apply appropriate retrieval strategies in Claude-based systems, including semantic search, keyword search, hybrid search, structured database queries, metadata-based retrieval, full-context retrieval, and choosing retrieval methods based on data structure and query patterns.	obj-17
18	Integration	Evaluate connection protocols for Claude-based integrations, including the Model Context Protocol (MCP), direct API integration, CLI-based integration, agent-to-agent (A2A) communication, choosing the appropriate protocol, understanding client/server responsibilities, tool discovery and execution, and authentication and permission boundaries.	obj-18
19	Integration	Evaluate the trade-offs between progressive discovery and monolithic context loading in Claude-based agent systems, including loading every tool upfront versus discovering tools on demand, progressive disclosure of capabilities, reducing context consumption, tool search and discovery, loading only relevant resources, and balancing discoverability, latency and complexity.	obj-19
20	Evaluation, Testing & Optimization	Define evaluation metrics covering accuracy, task completion rate, latency, cost per request, token consumption, reliability, safety, security, and user satisfaction.	obj-20
21	Evaluation, Testing & Optimization	Design evaluation datasets and frameworks, including building representative datasets, using real-world examples, edge cases,	obj-21

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		and adversarial test cases, creating regression datasets, using code-based and model-based graders, incorporating human evaluation, evaluating multi-turn conversations, and calibrating model graders against human judgment.	
22	Evaluation, Testing & Optimization	Conduct A/B testing and iterative improvement, including establishing a baseline, comparing model configurations, prompts, retrieval strategies, and alternative architectures, running experiments on controlled traffic, measuring statistical and practical improvements, and rolling back unsuccessful changes.	obj-22
23	Evaluation, Testing & Optimization	Diagnose system issues including prompt failures, hallucinations, incorrect retrieval, model mismatch, tool execution failures, context overflow, incorrect structured outputs, agent loops, and evaluation failures, and distinguish root causes from symptoms.	obj-23
24	Evaluation, Testing & Optimization	Optimize token usage, latency, and cost-performance using prompt caching, context reduction, model routing, batch processing, parallel execution, retrieval optimization, output token limits, model and effort selection, and cost per successful task.	obj-24
25	Evaluation, Testing & Optimization	Monitor performance in production through production evaluation, logging and metrics, distributed tracing, quality monitoring, regression detection, alerting, performance drift detection, and continuous improvement.	obj-25
26	Governance, Safety & Risk Management	Implement guardrails and safety controls spanning input validation and screening, output validation, prompt injection protection, jailbreak prevention, tool permission restrictions, system prompt guardrails, deterministic enforcement through code, monitoring and incident detection, and fail-open versus fail-closed behaviour.	obj-26
27	Governance, Safety & Risk Management	Identify risks, limitations and failure modes including hallucinations, biased outputs, prompt injection, indirect prompt injection through retrieved documents, sensitive data leakage, excessive permissions, unreliable agent behaviour, runaway agent execution, model deprecation and retirement, and context and token exhaustion.	obj-27
28	Governance, Safety & Risk Management	Apply human-in-the-loop validation, including mandatory approval before high-risk actions, human review after execution for reversible actions, confidence-based escalation, exception handling, human review queues, risk-based review strategies, and audit trails and approval records.	obj-28

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29	Governance, Safety & Risk Management	Ensure regulatory compliance across frameworks such as GDPR, HIPAA, and FedRAMP, covering data privacy and protection, data retention, data residency, access controls, auditability, deployment-specific compliance requirements, and how the chosen deployment platform affects compliance obligations.	obj-29
30	Governance, Safety & Risk Management	Address ethical AI considerations including bias detection, fairness, transparency, explainability, responsible AI, accountability, user disclosure, and monitoring discriminatory outcomes.	obj-30
31	Stakeholder Communication & Lifecycle Management	Conduct structured discovery and requirement gathering, including identifying business objectives, stakeholders, user workflows, functional and non-functional requirements, data sources, technical constraints, security and compliance requirements, business risks, and measurable success criteria.	obj-31
32	Stakeholder Communication & Lifecycle Management	Communicate architectural decisions and trade-offs through architecture decision records, technical design documentation, diagrams, cost/benefit explanations, risk communication to non-technical stakeholders, alternative comparisons, technology justification, and scalability/performance implications.	obj-32
33	Stakeholder Communication & Lifecycle Management	Manage stakeholder feedback and expectations by establishing feedback loops, defining SLAs, agreeing on measurable performance targets, communicating system limitations, managing expectations around probabilistic outputs, communicating model upgrades/changes, and translating feedback into improvements.	obj-33
34	Stakeholder Communication & Lifecycle Management	Document architectures and implementation guidance, including technical specifications, integration documentation, data flow diagrams, security documentation, API contracts, deployment instructions, operational runbooks, testing and evaluation documentation, and knowledge transfer.	obj-34
35	Stakeholder Communication & Lifecycle Management	Support the complete solution lifecycle, spanning discovery, architecture and design, implementation, testing and validation, deployment, monitoring, handover, continuous improvement, model migration, and decommissioning.	obj-35
36	Developer Productivity & Operational Enablement	Configure Claude Code and related tooling across organisation, project, and user levels — including managed settings, permissions, MCP servers, hooks, and shared project instructions — to support consistent, secure team environments.	obj-36
37	Developer Productivity &	Apply Claude Code commands, skills, and automation patterns to improve shared development workflows, including testing, code	obj-37

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	Operational Enablement	review, CI/CD integration, and measurable productivity gains across a team.	
38	Developer Productivity & Operational Enablement	Troubleshoot Claude Code configuration, MCP server failures, hooks, permissions, context issues, and tool execution failures, and monitor usage and operational behaviour to resolve developer-facing problems.	obj-38

Total: 640 practice questions across 38 pages.